PORTFOLIO

COMPUTER SCIENCE

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Software Developer · Age 22

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Selected work

1. Cosmic Observer
Windows astronomy desktop app — 3D sky, NASA data, AI tutor.

2. Vertical Farm Analysis
Python pipeline — telemetry to charts and PDF report.

> I build desktop applications and data pipelines with real deliverables: demos, screenshots, and client-ready reports.

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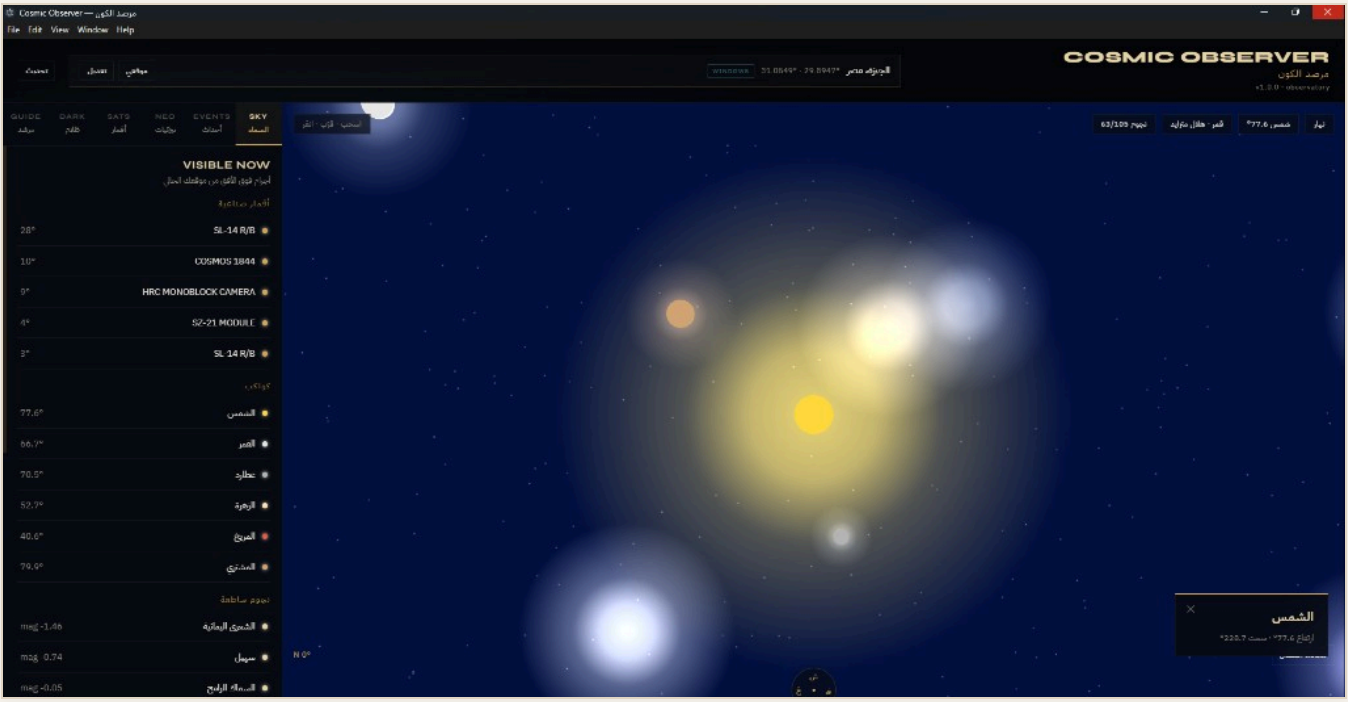
Cosmic Observer

Windows desktop astronomy app with interactive 3D sky, NEO tracking, orbital debris, dark-matter research views, and an AI tutor — shipped as a native **.exe**.



Capabilities

- Live 3D sky from GPS or manual location
- Satellites, planets, and bright stars
- Sun / Moon status and sky events
- Modules: Sky, Events, NEO, Sats, Dark, Guide
- Arabic + English interface

Links

- github.com/engboda6/cosmic-observer
- License: MIT
- Platform: Windows (.exe)
- Demo video on portfolio site

Stack & Features

JavaScript end to end — React + Three.js for the sky, Node.js/Express for APIs, Electron for the Windows desktop shell.

JavaScript | React 18 | Three.js | Vite | Node.js | Express | astronomy-engine | OpenAI | Electron 33 | GLSL

Features

- Interactive 3D celestial sphere
- Sunrise / sunset and moon phases
- NASA NeoWs near-Earth objects
- CelesTrak debris / TLE tracking
- Dark-matter lensing research views
- Optional OpenAI astronomy tutor

Data sources

- NASA NeoWs
- JPL Sentry / CAD
- NASA CNEOS Fireball
- CelesTrak (TLE)
- VizieR / CDS

Structure

- `electron/` — desktop shell
- `server/` — local APIs
- `src/` — React + Three.js

Vertical Farm Analysis

Pure Python pipeline for 600 telemetry rows. Report prepared by **Abd Alrahman Ahmed**.

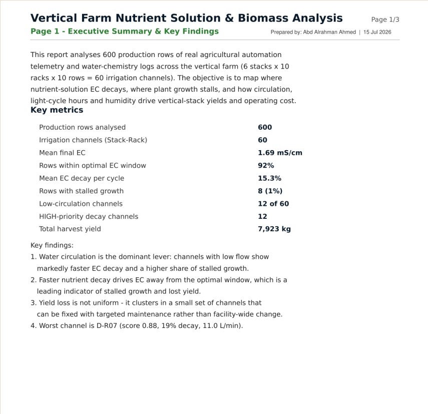
1 **prepare_dataset.py** — clean CSV and parse Stack / Rack / Row IDs

2 **analysis.py** — decay matrix for 60 channels, harvest index, feeding balances

3 **charts.py** — EC distribution, heatmap, harvest charts

4 **report.py** — three-page PDF for delivery

5 **run_pipeline.py** — full run in one command



Vertical Farm Nutrient Solution & Biomass Analysis Page 1/3

Page 1 - Executive Summary & Key Findings — Prepared by: Abd Alrahman Ahmed | 15 Jul 2026

This report analyses 600 production rows of real agricultural automation telemetry and water-chemistry logs across the vertical farm (6 stacks x 10 racks x 10 rows = 60 irrigation channels). The objective is to map where nutrient-solution EC decays, where plant growth stalls, and how circulation, light-cycle hours and humidity drive vertical-stack yields and operating cost.

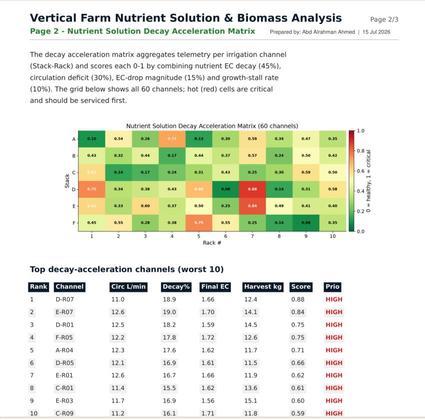
Key metrics

Production rows analysed	**600**
Irrigation channels (Stack-Rack)	**60**
Mean final EC	**1.69 mS/cm**
Rows within optimal EC window	**92%**
Mean EC decay per cycle	**15.3%**
Rows with stalled growth	**8 (1%)**
Low-circulation channels	**12 of 60**
HIGH-priority decay channels	**12**
Total harvest yield	**7,923 kg**

Key findings:

1. Water circulation is the dominant lever: channels with low flow show markedly faster EC decay and a higher share of stalled growth.
2. Faster nutrient decay drives EC away from the optimal window, which is a leading indicator of stalled growth and lost yield.
3. Yield loss is not uniform - it clusters in a small set of channels that can be fixed with targeted maintenance rather than facility-wide change.
4. Worst channel is D-R07 (score 0.88, 19% decay, 11.0 L/min).

Page 1 — Summary



Vertical Farm Nutrient Solution & Biomass Analysis Page 2/3

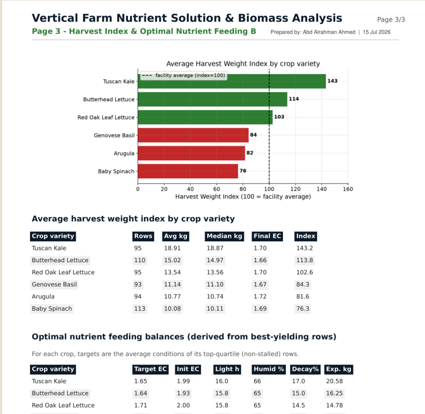
Page 2 - Nutrient Solution Decay Acceleration Matrix — Prepared by: Abd Alrahman Ahmed | 15 Jul 2026

The decay acceleration matrix aggregates telemetry per irrigation channel (Stack-Rack) and scores each 0-1 by combining nutrient EC decay (45%), circulation deficit (30%), EC-drop magnitude (15%) and growth-stall rate (10%). The grid below shows all 60 channels; hot (red) cells are critical and should be serviced first.

Top decay-acceleration channels (worst 10)

Rank	Channel	Circ L/min	Decay%	Final EC	Harvest kg	Score	Prio
1	D-R07	11.0	18.9	1.66	12.4	0.88	HIGH
2	E-R07	12.6	19.0	1.70	14.1	0.84	HIGH
3	D-R01	12.5	18.2	1.59	14.5	0.75	HIGH
4	F-R05	12.2	17.8	1.72	12.6	0.75	HIGH
5	A-R04	12.3	17.6	1.62	11.7	0.71	HIGH
6	D-R05	12.1	16.9	1.61	11.5	0.66	HIGH
7	E-R01	12.6	16.7	1.66	11.9	0.62	HIGH
8	C-R01	11.4	15.5	1.62	13.6	0.61	HIGH
9	E-R03	11.7	16.9	1.56	15.1	0.60	HIGH
10	C-R09	11.2	16.1	1.71	11.8	0.59	HIGH

Page 2 — Matrix



Vertical Farm Nutrient Solution & Biomass Analysis Page 3/3

Page 3 - Harvest Index & Optimal Nutrient Feeding B — Prepared by: Abd Alrahman Ahmed | 15 Jul 2026

Average harvest weight index by crop variety

Crop variety	Rows	Avg kg	Median kg	Final EC	Index
Tuscan Kale	95	18.91	18.87	1.70	143.2
Butterhead Lettuce	110	15.02	14.97	1.66	113.8
Red Oak Leaf Lettuce	95	13.54	13.56	1.70	102.6
Genovese Basil	93	11.14	11.10	1.67	84.3
Arugula	94	10.77	10.74	1.72	81.6
Baby Spinach	113	10.08	10.11	1.69	76.3

Optimal nutrient feeding balances (derived from best-yielding rows)

For each crop, targets are the average conditions of its top-quartile (non-stalled) rows.

Crop variety	Target EC	Init EC	Light h	Humid %	Decay%	Exp. kg
Tuscan Kale	1.65	1.99	16.0	66	17.0	20.58
Butterhead Lettuce	1.64	1.93	15.8	65	15.0	16.25
Red Oak Leaf Lettuce	1.71	2.00	15.8	65	14.5	14.78

Page 3 — Harvest

Libraries & Contact

Raw farm CSV becomes a clean dataset, analysis tables, charts, Excel files, and a three-page report.

Libraries

- **pandas** — cleaning & aggregation
- **numpy** — calculations
- **matplotlib** — charts & PDF figures
- **openpyxl** — Excel export

Deliverables

- Clean structured dataset
- Analysis for 60 irrigation channels
- Charts and Excel workbooks
- Three-page nutrient PDF

Contact

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